

Test Sample Paper B

Total Score
100



Name: _____

Section: _____ Classes held on: _____

INSTRUCTIONS TO STUDENTS

1. Write your full name, section number and timing of your classes in the space provided above.
2. This paper contains **2** sections. Section A has 12 questions of 3 marks each. Section B contains questions 13-30 and the marks are stated next to the each question.
3. You are required to answer **ALL** questions. Write your answer in the space provided after each question. If applicable, **write your workings legibly** in the space provided after each question. No separate papers are allowed.
4. For multiple choice questions, choose the **MOST APPROPRIATE** answer from the alternatives given. Multiple answers are **not** applicable for this paper.
5. This test constitutes 20% of your total assessment of this MA module.
6. Time allowed is 90 minutes.

Section A: MCQ

1. Which of the following is **not** a characteristic of management accounting information?
- A. Emphasizes on timeliness more than precision.
 - B. Focuses on the future more than the past.
 - C. Emphasizes verifiability more than relevance.
 - D. Has flexible formats rather than having to following the international accounting standards.
 - E. Provides detailed information about parts of the company, not just the company as a whole.

Answer: [_____]

2. Which of the following(s) would affect the classification of direct versus indirect costs?
- A. The cost object.
 - B. The materiality of the costs.
 - C. Whether the cost can be easily and conveniently traced to a cost object.
 - D. Availability of information-gathering technology.
 - E. All of the above.

Answer: [_____]

3. The advantages of the high-low method of cost estimation include:
- I. only limited data is required
 - II. for a very accurate projection
 - III. if all data follows the linear function well, the method can give quick and relevant data for decision making
- A. I only
 - B. I and II only
 - C. II and III only
 - D. I and III only
 - E. I, II and III

Answer: [_____]

4. Fix-IT is a new company which intends to help other companies fix their computers. Sales revenue will be earned at a rate of S\$100 per hour worked on each computer. Total fixed costs per month (excluding salaries) are estimated to be S\$50,000. There is no variable cost.

Fix-IT has a choice between

Option 1: Hiring permanent IT staff at S\$50,000 per month to fix the computers in Singapore and

Option 2: Outsourcing all the work to a company based in Malaysia. The Malaysian company will charge Fix-IT S\$40 per hour worked on each computer, but it will cost Fix-IT an additional S\$5,000 per month in fixed costs to transport computers between Singapore and Malaysia.

Which of the following statements is **true** for any one month?

- A. Option 1 has a higher contribution margin ratio.
- B. Option 1 has a higher breakeven point.
- C. Option 1 will result in Fix-IT having a lower operating leverage.
- D. Both "A" and "B" are true.
- E. All of the above are true.

Answer: [_____]

5. Which of the following statements relating to variable costing and absorption costing is **false**?

- A. The product costs for absorption costing differ from that of variable costing in that the product costs for absorption costing includes fixed selling and administrative expenses.
- B. Variable costing is consistent with the contribution margin format.
- C. Under variable costing, contribution margin is after deducting variable cost of goods sold and variable selling and administrative costs from sales revenue.
- D. When finished goods inventory decreases during a period, a manufacturing company's absorption costing net operating income for that period will usually be smaller than its variable costing net operating income.
- E. Regardless of the sales volume versus production volume, as long as a company has ending inventory, the company's retained profit (cumulative profit from day 1) will always be higher under absorption costing than under variable costing.

Answer: [_____]

Use the following information for questions 6-7:

Hamstein Semiconductors has the following information in its Photolithography Department during December.

Materials are added at the beginning of the photolithography process.

Units:	
Work in process, December 1	7,000 units
Transferred-in from the Polishing and Cutting Department during December	29,000 units
Work in process, December 31 (70% completed)	11,000 units
Costs:	
Work in process, December 1	\$65,550
Transferred-in from the Polishing and Cutting Department during December	\$97,300
Direct materials added during December	\$69,700
Conversion costs added during December	\$90,700

The work in process costs as at December 1 pertained to the following:

Transferred-in costs	\$21,500
Direct materials costs	\$20,300
Conversion costs	<u>\$23,750</u>
	<u>\$65,550</u>

6. Total equivalent units of production through the conversion process during December in the Photolithography Department are:
- A. 25,000
 - B. 32,700
 - C. 36,000
 - D. 36,700
 - E. None of the above

Answer: [_____]

7. Total costs of the completed and transferred-out units in the Photolithography Department are:
- A. \$90,750
 - B. \$150,000
 - C. \$232,500
 - D. \$241,817
 - E. None of the above

Answer: [_____]

8. For performance evaluation purposes, the variable costs of a service department should be charged to operating departments using:
- A. actual total variable costs for the period.
 - B. budgeted total variable costs for the period.
 - C. budgeted level of services times budgeted variable rate per service.
 - D. actual level of services consumed times budgeted variable rate per service.
 - E. None of the above.

Answer: [_____]

9. Which of the following statements is (are) **true** regarding Job Costing?
- I. Direct costs are traced to the jobs and indirect costs are lumped together and allocated to the jobs.
 - II. Companies typically wait for accurate information regarding actual overhead costs before pricing a job.
 - III. Job costing can be used for both manufacturing and service companies.
- A. I only
 - B. I and II
 - C. I and III
 - D. II and III
 - E. I and II and III

Answer: [_____]

10. Which of the following products/services cost is **most likely** to be accounted for using job costing rather than process costing?

- A. Production of orange juice at Sunkist.
- B. Manufacturing of potato chips at Lay's.
- C. Provision of tax preparation service for corporate clients at Ernst and Young.
- D. Provision of customer support service at a call center.
- E. Production of jeans at Levi's.

Answer: [_____]

11. Which of the following statements is **true**?

- A. Organization-sustaining activities are activities of the organization that support specific products.
- B. Activity-based costing is a costing method that is designed to provide managers with product cost information for external financial reports.
- C. Product costs calculated based on activity-based costing is always higher than the product costs calculated based on traditional costing because activity-based costing also allocates non-manufacturing costs to products.
- D. Unit-level activities are performed each time a unit is produced.
- E. The costs of a particular department should not be split up among activity cost pools in an activity-based costing system.

Answer: [_____]

12. Consider the following activities relating to a product:

- I. Research and Development
- II. Advertising
- III. Packaging the units of products

Which of the above activity (or activities) is (are) example(s) of product-level activities?

- A. I only.
- B. II only.
- C. III only.
- D. I and II.
- E. II and III.

Answer: [_____]

SECTION B: Structured Questions (64 marks total)

13. Using the high-low method and the data provided below, calculate the variable cost % of each sales \$. (Round to the nearest %). Assume no change in selling price and cost price during the five months and the company does not keep inventory.

Month	Total Mixed Cost (\$)	Sales \$
January	37,000	370,000
February	47,000	490,000
March	41,000	420,000
April	16,995	120,500
May	17,000	120,000

Answer: [_____]
(2 marks)

14. Hotel 88 has two types of rooms available: 400 standard rooms and 100 luxury rooms. The price of each room and housekeeping costs are as follow:

	Standard Room (\$)	Luxury Room (\$)
Price per room per night	100	200
Housekeeping cost per room per night	(50)	(75)

The total annual fixed cost of running the hotel is \$8,905,000. What is the breakeven number of room nights for the standard rooms? (Note: one "room night" is one room occupied in one night.)

Answer: [_____]
(4 marks)

15. A manufacturing company has incurred the following costs and inventory balances during its recent financial year.

<u>Expenses</u>	\$
Purchase of direct materials	500,000
Direct labour	600,000
Depreciation for the manufacturing machines	500,000
Depreciation for the office equipment	10,000
Utility costs (80% for factory and 20% for office)	100,000
Other factory expenses	220,000
Marketing expenses	250,000
Distribution costs	50,000
Accountancy and legal fees	55,000
Other selling and administrative expenses	80,000

<u>Balance Sheet - Inventory</u>	\$
Beginning direct materials	300,000
Ending direct materials	200,000
Beginning work-in-process	400,000
Ending work-in-process	100,000
Beginning finished goods	50,000
Ending finished goods	50,000

What is the company's cost of goods manufactured? Assume that all manufacturing costs are treated as product costs.

Answer: [_____]
(4 marks)

16. Super Donuts Limited makes and sells a variety of donuts. In 20X1, the company had the following results:

	\$
Sales	2,600,100
Variable Costs: Donut ingredients	520,020
Total Fixed Costs (rent, wages, depreciation etc)	1,100,000
Operating profit	980,080

The following information has been gathered regarding 20X2:

- No change in selling price of the donuts.
- Cost of donut ingredients will go up by 10% because of less food supply due to bad weather in many countries.
- Except donut ingredients which are variable costs, all other costs are fixed.
- Due to a new food safety regulation, the company will incur an addition wages of \$36,000 for the hiring of a food inspector.

Based on the above information, compute the required sales dollars to be made in 20X2 in order to achieve the same operating profit as in 20X1, round your answer to the nearest dollar.

Answer: [_____]
(4 marks)

17. The following per call data apply to the current “normal” call volume of 10,000 calls per day:

	INR
Variable wages	20
Variable overhead	3
Fixed overhead	10
Total cost per call	33

The call centre has capacity for 30,000 calls per day and is considering expanding its service to 20,000 calls per day by serving more companies. What is the total cost of doing 20,000 calls per day?

Answer: [_____]
(2 marks)

Use the following information for question 18-21:

YWC Corporation uses a predetermined fixed overhead rate of \$4 per unit.

For the month of June 20X2, YWC Corporation had the following data:

Units manufactured	25,000
Units sold	20,000
Units in beginning inventory	0
Selling price per unit	\$20
Product costs:	
Variable per unit	\$12
Total Fixed	\$100,000
Selling and Administrative:	
Variable per unit	\$1
Total Fixed	\$15,000

18. Using the variable costing format, prepare an income statement showing clearly the figures for variable cost of goods sold, contribution margin and operating profit.
(4 marks)
19. What would have been the operating profit using the absorption income statement format?
(4 marks)
20. Reconcile the difference in variable operating income and absorption operating income computed in the previous two questions based on a production level of 25,000 units in the month. Briefly explain the difference.
(2 marks)
21. If sales and costs are unchanged, what can YWC do to increase operating profit by \$5,000? Assume YWC writes off under-/over-applied overhead to cost of goods sold.
(2 marks)

Answers for questions 18 to 21

Use the following information for questions 22-23:

Rich Company has two service departments, Custodial Services and Maintenance department as well as two operating departments, Machinery and Milling. Costs of the service departments are allocated to the operating departments on the basis of square metres of space occupied.

In January, the amount of space occupied by each department was:

Custodial services	6,000 square metres
Maintenance	9,000 square metres
Machinery	45,000 square metres
Milling	90,000 square metres

Actual departmental costs in Custodial Services and Maintenance department were \$86,400 and \$89,100 respectively for January. The company uses the

22. Calculate the amount of Maintenance department cost allocated to Milling if the company uses **direct method of service cost allocation**.

Answer: [_____]
(2 marks)

23. Calculate the amount of Maintenance department cost allocated to Milling if the company uses **Step-down method of service cost allocation** starting from Custodial services.

Answer: [_____]
(4 marks)

Use the following information for questions 24-26:

Warner Company currently allocates its manufacturing overheads to 3 products (Regular, Slim Fit and Customize) based on the traditional method. The profitability of the company's 3 products for year 20X3 is given in the table below:

	Regular	Slim Fit	Customize
Sales Units	25,000	30,000	8,000
Per unit			
Selling price	\$400	\$400	\$640
Direct material cost	\$120	\$180	\$300
Direct labour cost	\$80	\$100	\$125
Manufacturing overheads	\$180	\$130	\$200
Profit/(Loss)	\$20	(\$10)	\$15

All the direct labor workers have similar skills and cost on average, \$20 per direct labor hour. Direct material, a special fabric, is used for all the company's products. The fabric costs \$60 per square meter (sqm).

The company plans to implement activity-based costing to allocate the manufacturing overheads (MOH) to the 3 product lines based on the following:

- 9% allocated by direct labor hours
- 18% driven by the number of square meters of fabric used
- 23% based on batch related activities. Batch sizes of the three products, Regular, Slim Fit and Customize are 250 units, 100 units and 80 units respectively.
- 50% are allocated by design changes. There were 70 design changes for Regular, 10 for Slim Fit and 120 for Customize.

24. Using the activity-based costing method, complete the following table.

(4 marks)

	Activity Levels			
Activity Drivers	(A) Regular	(B) Slim Fit	(C) Customize	(D) =A+B+C Total
No. of direct labor hours				
No. of sqm of direct materials used				
No. of batches				
No. of design changes	70	10	120	200

25. Compute the activity rates using the following table.

(4 marks)

	(E)	(F)	
	% of MOH	Activity Cost Pool (\$)	Activity Rates (\$)
No. of direct labor hours	9%		
No. of sqm of direct materials used	18%		
No. of batches	23%		
No. of design changes	50%		
(F) Total Manufacturing Overhead (MOH \$)	100%		

26. Re-compute the total profit or loss for Slim Fit using the activity-based costing system.

Answer: [_____]

(4 marks)

Use the following information for questions 19 to 20

Three SMU graduates have just set up a website design firm in January 20X4. Based on the graduates' knowledge of Management Accounting, they plan to use a **normal costing system** and trace direct costs to each job. Indirect costs are allocated to jobs based on a predetermined indirect cost allocation rate computed as a percentage of professional labour costs. All under- or over-applied indirect costs are written off to the cost of services sold.

For the first year of operation, the three partners budgeted \$180,000 as the total annual professional labour costs (direct costs) and \$82,800 as indirect costs based on the following items:

Support staff salaries	\$24,000
Office rent	\$36,000
Depreciation of computers and equipment	\$5,000
Office supplies	\$5,800
Other indirect costs	\$12,000
Total indirect costs	\$82,800

During January 20X4, the firm started three jobs, two jobs were completed and sold and one job was still in progress at the end of January 20X4, details of the direct costs of the three jobs are as follows:

	Job A	Job B	Job C
	Completed and Sold	Completed and Sold	Work in Progress
Professional labour costs	\$8,300	\$2,500	\$5,100
Licensing costs	\$230	\$0	\$400
Travel costs	\$50	\$30	\$300

For January 20X4, actual total professional labour costs and actual total indirect costs were \$15,900 and \$5,800 respectively.

27. Compute (i) the predetermined overhead rate for year 20X4 and (ii) the total cost of services sold for January 20X4 before the adjustment of under- or over-applied indirect costs.

Answer (i): [_____]
(2 marks)

Answer (ii): [_____]
(4 marks)

28. Compute (i) the under- or over-applied indirect costs for January 20X4 and (ii) the adjusted total cost of services sold for January 20X4.

Answer (i): [_____]
Under or over (delete one) applied
(2 marks)

Answer (ii): [_____]
(2 marks)

Use the following information for questions 29-30:

Rocket Pizza is a manufacturer of frozen microwavable pizzas to supermarkets in Singapore. The manufacturing of frozen pizzas involves three processes across various departments. The first process (dough-making) is the preparation of the pizza dough, where flour, water, oil, yeast, salt and sugar are mixed together by machine. The dough is then left to sit until it rises to twice its size. The finished dough is then cut into individual dough balls and transferred to the second department (baking) where the ingredients are added to the dough and the pizzas are baked. Finally in the third department (packaging), the finished pizzas are flash-frozen and packed in boxes to be delivered to customers.

At the beginning of March, the cost information pertaining to the beginning work in process in the second department (baking) is as follows:

Work-in-process, March 1	5,000 pizzas
Transferred-in material (dough balls)	\$4,750
Sauce and ingredients	\$17,500
Cheese	\$2,800
Conversion costs	\$4,880

The manufacturing process in the second department (baking) is described as follows. At the beginning of the process, the dough balls are flattened into round-shaped pizzas. 60% of the sauce and ingredients are then added to the pizzas before they are transferred to the oven for baking. When the baking process is 50% completed, the remaining sauce and ingredients are added. At the same time, 80% of the cheese is added. The remaining cheese is then added when the process is 90% completed. Conversion costs are assumed to be added uniformly throughout the entire production process.

In March, 25,000 dough balls are transferred from the dough-making department into the baking department. During the month of March, the following costs are incurred in the baking department:

Transferred-in material (dough balls)	\$25,250
Sauce and ingredients	\$83,240
Cheese	\$19,600
Conversion costs	\$31,120

At the end of March, 2,000 units of unfinished pizzas were 40% completed in the baking department. Rocket Pizza uses the weighted average cost method.

29. Compute the equivalent units of the following raw materials in the ending work-in-process (EWIP) inventory:

- i) Sauce and ingredients
- ii) Cheese.

Answer (i): [_____]
(2 marks)

Answer (ii): [_____]
(2 marks)

30. Compute the cost of ending work-in-process in the baking department at the end of March.

Answer: [_____]
(4 marks)

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End of the Test